



UNITED INTERNATIONAL UNIVERSITY (UIU)
DIGITAL SIGNAL PROCESSING LABORATORY
COURSE CODE EEE 3310

Lab Report : 04

Frequency Domain Analysis using MATLAB

Group No: 02

Group Members:

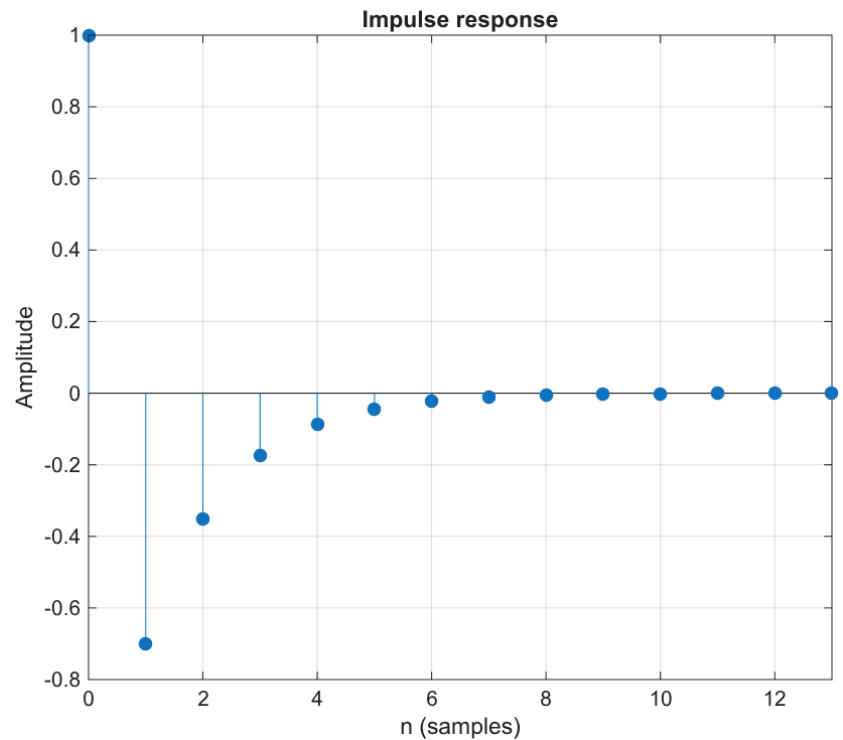
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Home Work Answers:

Q1: (i)

Matlab Code:

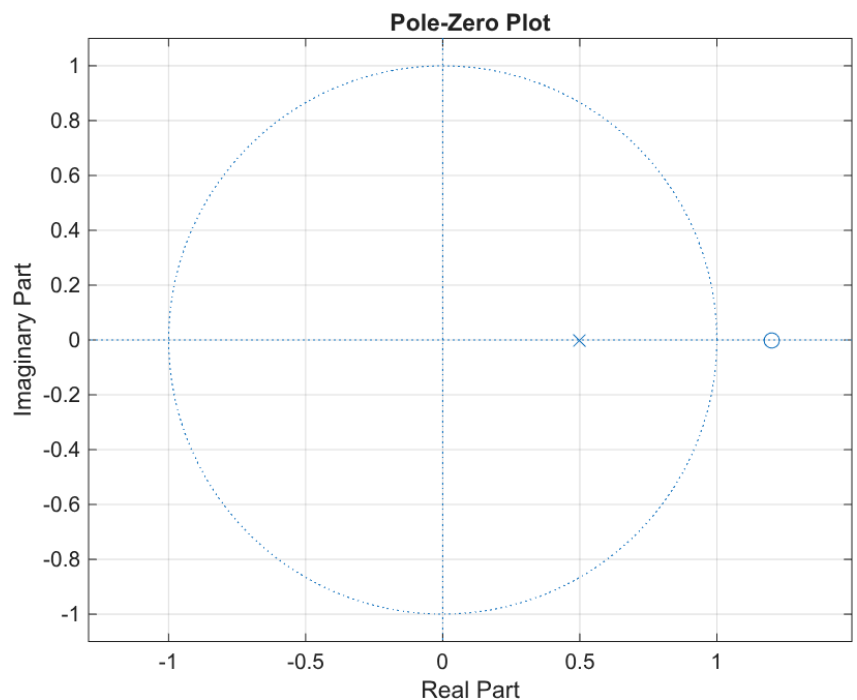
```
%% Q1
clc; close all; clear;
b=[1 -1.2];
a=[1 -0.5];
%(i)
figure
impz(b,a)
title('Impulse response')
grid on;
```



The impulse response decays exponentially, so the system is stable.

Q1: (ii)

```
clc; close all; clear;
b=[1 -1.2];
a=[1 -0.5];
%(ii)
figure
zplane(b,a)
title('Pole-Zero Plot')
grid on;
```

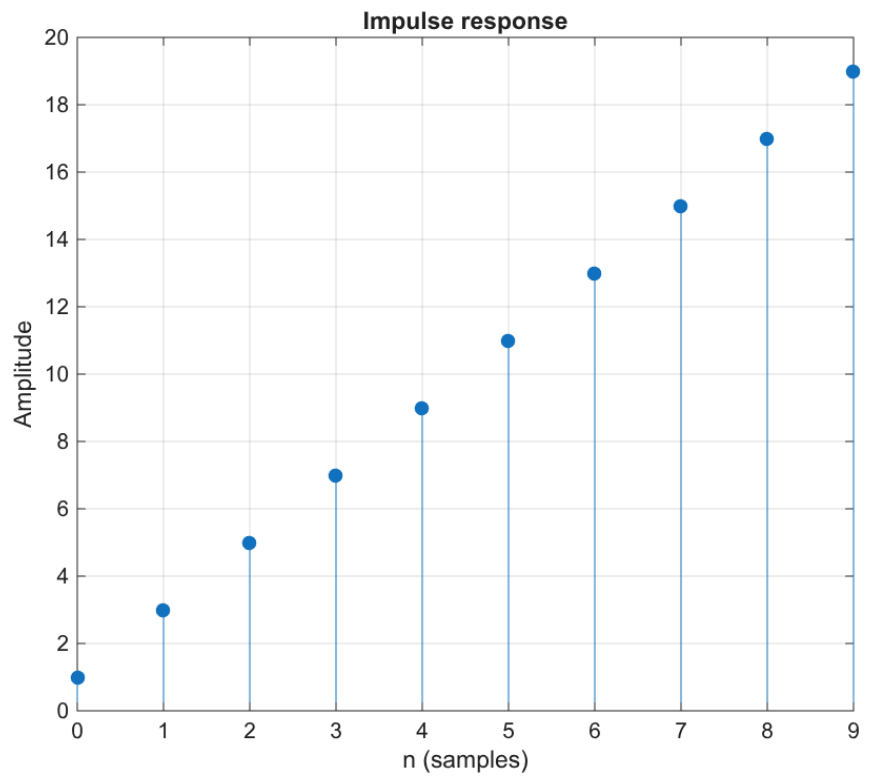


Pole is inside unit circle, so its a stable system.

Q2: (i)

Matlab Code:

```
%% Q2
clc; close all; clear;
b=[1 1];
a=[1 -2 1];
%(i)
figure
impz(b,a)
title('Impulse response')
grid on;
```



The impulse response grows linearly, so the system is unstable.

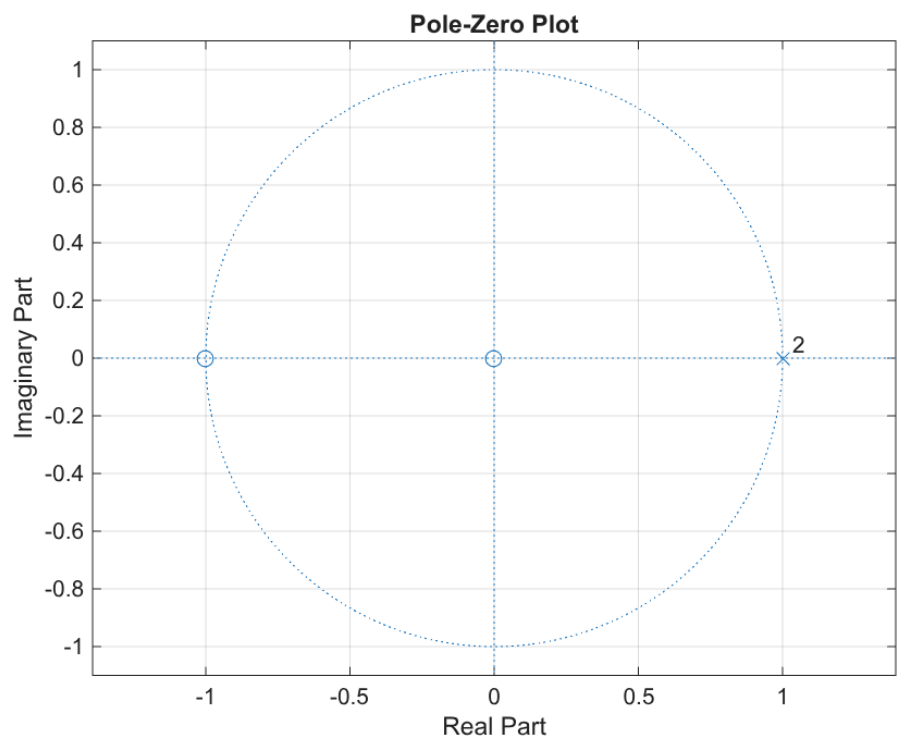
Q2: (ii)

Multiple(2) poles on the unit circle, so the system is unstable.

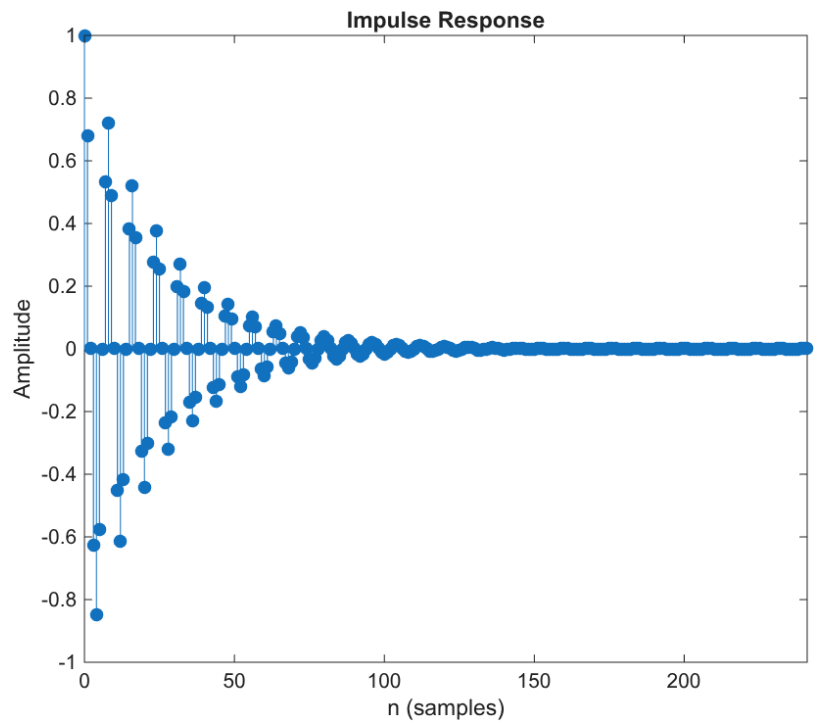
Matlab Code:

```
%% Q2
clc; close all; clear;
b=[1 1];
a=[1 -2 1];

%(ii)
figure
zplane(b,a)
title('Pole-Zero Plot')
grid on;
```



Q3: (a)



Matlab Code:

%% Q3

clc; clear all; close all;

% (a)

r1 = 0.96;

b1 = [1, -0.707*r1];

a1 = [1, -sqrt(2)*r1, r1^2];

impz(b1, a1);

Dependence on r:

- r^n an exponential decay factor
- If $r = 0.96$, decay is very slow
- $r = 0.71$, decay is medium
- $r = 0.14$, decay is extremely fast

Q:3

$$p_{1,2} = re^{\pm j\pi/4} \quad z_1 = 0.707r = \frac{1}{\sqrt{2}}r$$

$$= r(\cos \frac{\pi}{4} \pm j \sin \frac{\pi}{4})$$

$$= r(\frac{1}{\sqrt{2}} \pm j \frac{1}{\sqrt{2}})$$

$$= r \frac{1}{\sqrt{2}} (1 \pm j)$$

Transfer function $\rightarrow H(z) = \frac{1 - z_1 z^{-1}}{(1 - p_1 z^{-1})(1 - p_2 z^{-1})} = \frac{1 - \frac{r}{\sqrt{2}} z^{-1}}{1 - \sqrt{2}}$

$$= \frac{1 - z_1 z^{-1}}{1 - p_2 z^{-1} - p_1 z^{-1} + p_1 p_2 z^{-2}}$$

$$= \frac{1 - z_1 z^{-1}}{1 - z^{-1}(p_1 + p_2) + p_1 p_2 z^{-2}} \quad | p_{1,2} = re^{\pm j\pi/4}$$

$$H(z) \approx \frac{1 - \frac{r}{\sqrt{2}} z^{-1}}{1 - \sqrt{2} r z^{-1} + r^2 z^{-2}} \quad \left| \begin{array}{l} b_0 = [1 \quad -\frac{r}{\sqrt{2}}] \\ a_2 = [1 \quad \sqrt{2} \quad r^2] \end{array} \right.$$

$$h(n) = r^n \cos\left(\frac{\pi}{4}n\right) u(n)$$

$$H(z) = \frac{1 - (r \cos \frac{\pi}{4}) z^{-1}}{1 - (2r \cos \frac{\pi}{4}) z^{-1} + r^2 z^{-2}}$$

$$\downarrow$$

$$h(n) = r^n \cos\left(\frac{\pi}{4}n\right) u(n)$$

Q3: (b)

```
clc; clear all; close all;
```

```
% (b)
```

```
r1 = 0.96;
```

```
b1 = [1 -0.707*r1];
```

```
a1 = [1 -sqrt(2)*r1 r1^2];
```

```
[H1,w]=freqz(b1,a1,100);
```

```
magH1=abs(H1);
```

```
phaH1=angle(H1);
```

```
figure;
```

```
subplot(2,1,1); plot(w/pi,magH1);
```

```
grid;
```

```
title('Magnitude Response');
```

```
xlabel('frequency in pi units');
```

```
ylabel('Magnitude');
```

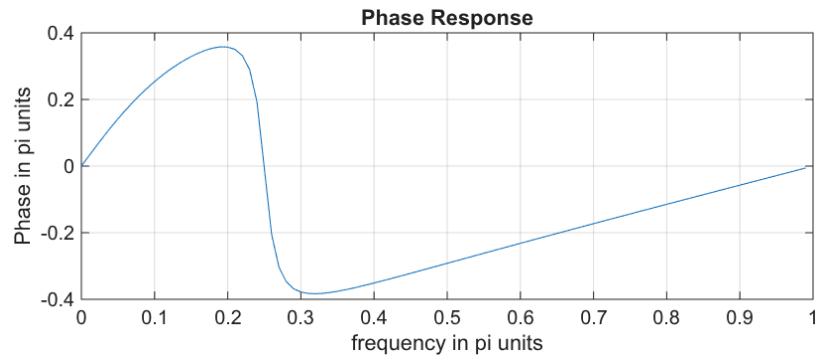
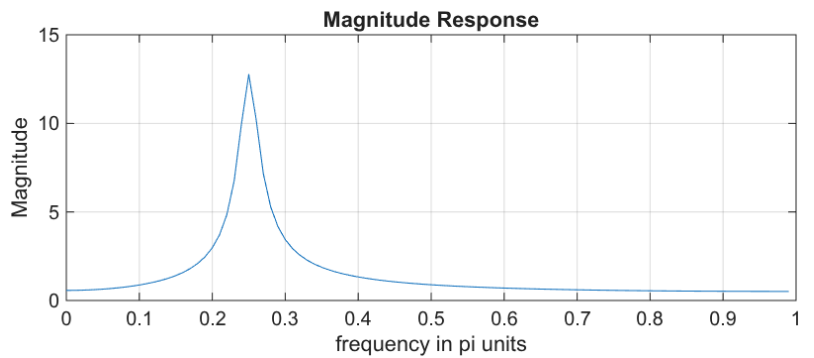
```
subplot(2,1,2); plot(w/pi, phaH1/pi);
```

```
grid;
```

```
title('Phase Response');
```

```
xlabel('frequency in pi units');
```

```
ylabel('Phase in pi units');
```



Q3: (c)

```
%% (c)
```

```
r2=0.71;
```

```
b2=[1 -0.707*r2];
```

```
a2=[1 -sqrt(2)*r2 r2^2];
```

```
[H2,w]=freqz(b2,a2,100);
```

```
magH2=abs(H2);
```

```
phaH2=angle(H2);
```

```
figure;
```

```
subplot(2,1,1); plot(w/pi,magH2);
```

```
grid;
```

```
title('Magnitude Response');
```

```
xlabel('frequency in pi units');
```

```
ylabel('Magnitude');
```

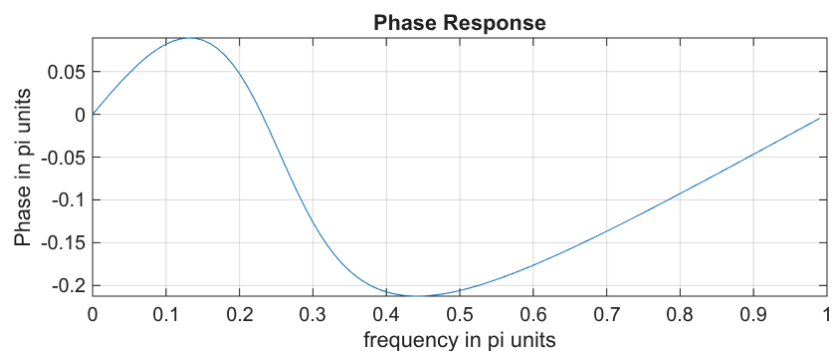
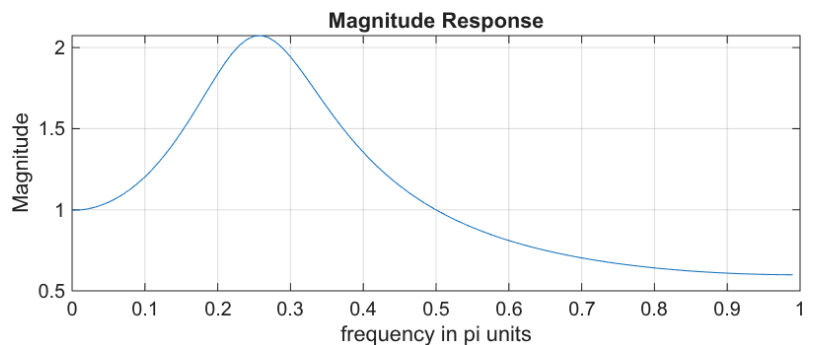
```
subplot(2,1,2); plot(w/pi, phaH2/pi);
```

```
grid;
```

```
title('Phase Response');
```

```
xlabel('frequency in pi units');
```

```
ylabel('Phase in pi units');
```



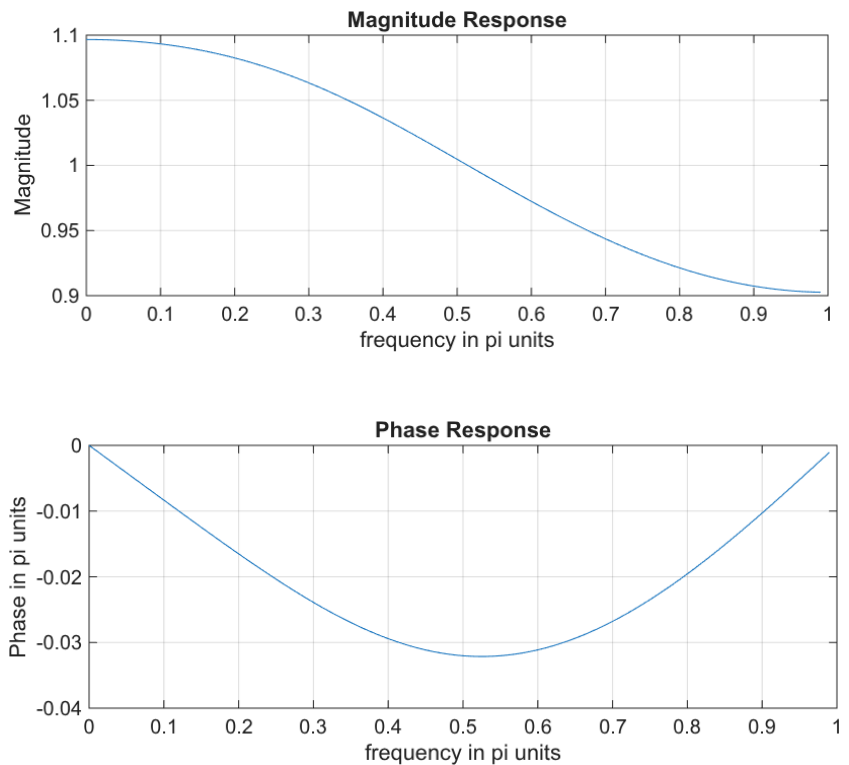
Q3: (d)

%% (d)

```
r3=0.14;  
b3=[1 -0.707*r3];  
a3 = [1 -sqrt(2)*r3 r3^2];
```

```
[H3,w]=freqz(b3,a3,100);  
magH3=abs(H3);  
phaH3=angle(H3);
```

```
figure;  
subplot(2,1,1); plot(w/pi, magH3);  
grid;  
title('Magnitude Response');  
xlabel('frequency in pi units');  
ylabel('Magnitude');  
  
subplot(2,1,2); plot(w/pi, phaH3/pi);  
grid;  
title('Phase Response');  
xlabel('frequency in pi units');  
ylabel('Phase in pi units');
```



Q3: (e)

When r gets smaller (from 0.96 down to 0.14), the sharp peak at 0.25π flattens out, and the signal strength drops. The phase line becomes smoother and flatter.

When r is close to 1 (closer to the unit circle), the filter becomes much better at picking out its main frequency.

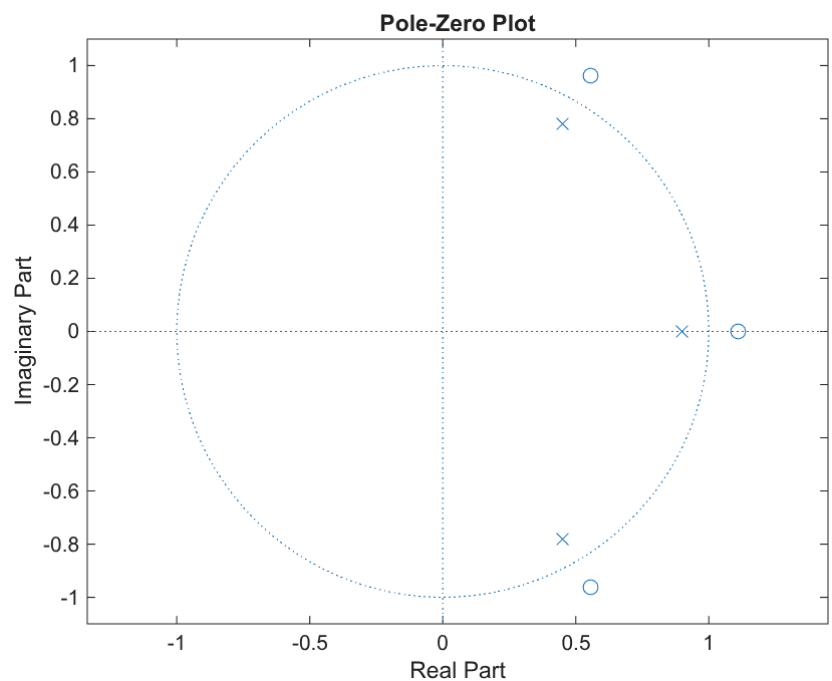
Q4: (a)

%% Q4

```
clc; clear all; close all;
```

```
b=[-0.729 1.62 -1.8 1];  
a=[1 -1.8 1.62 -0.729];
```

```
zplane(b,a)
```



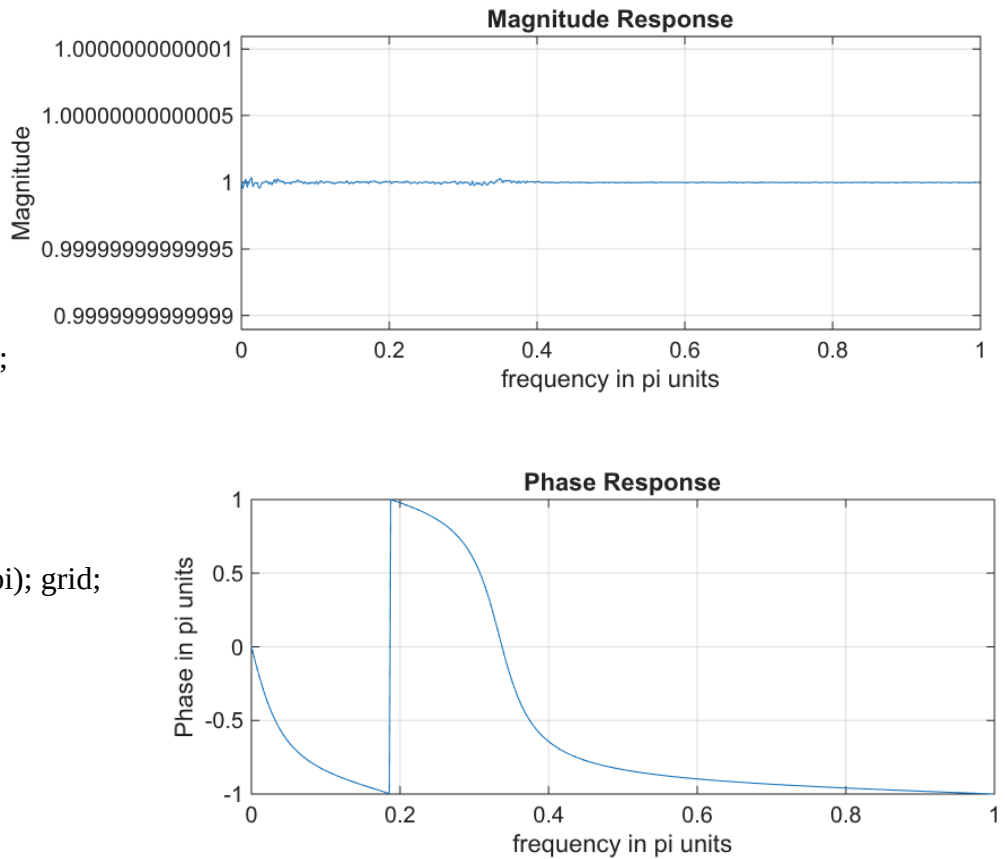
Q4: (b)

```
clc; clear all; close all;  
b=[-0.729 1.62 -1.8 1];  
a=[1 -1.8 1.62 -0.729];
```

```
[H,w]=freqz(b,a,512);  
magH=abs(H);  
phaH=angle(H);
```

```
figure;  
subplot(2,1,1); plot(w/pi,magH);  
grid;  
title('Magnitude Response');  
xlabel('frequency in pi units');  
ylabel('Magnitude');
```

```
subplot(2,1,2); plot(w/pi,phaH/pi); grid;  
title('Phase Response');  
xlabel('frequency in pi units');  
ylabel('Phase in pi units');
```



Q4: (C)

The magnitude looks constant and equal to 1 across all frequencies from 0 to π . It does not amplify any frequency component.

The phase decreases non-linearly from 0 to negative values as frequency increases, which means it delays different frequencies by different amounts.

Q4: (d)

This system is an All-Pass Filter.

It lets all frequencies pass through without changing their strength (gain stays at 1), and only changes their timing/phase.

Its zeros are placed inverse to its poles across the unit circle.